METHOD AND SYSTEM FOR GEOGRAPHICAL AND ORIENTATION-BASED ENCRYPTION SEEDS

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Title: METHOD AND SYSTEM FOR GEOGRAPHICAL AND ORIENTATION-BASED ENCRYPTION SEEDS

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Abstract

A method and system for generating cryptographic keys using dynamic, real-time device orientation and geographical location data as entropy sources. The method comprises acquiring precise geographical coordinates from a Global Navigation Satellite System (GNSS or GPS) receiver and device orientation data from an internal digital compass or inertial measurement unit (IMU). These highly variable and difficult-to-predict environmental parameters are then combined and processed through a cryptographic hashing function to generate unique encryption seeds. This approach significantly enhances the security of communication channels by providing high-fidelity, continuously changing entropy that is challenging for state-sponsored malicious actors to predict or replicate, even under physical surveillance, unlike time-based seeds.

Background of the Invention

Traditional cryptographic key generation often relies on entropy sources such as timestamps, random number generators, or other system-level variables. While effective against many forms of attack, time-based seeds, in particular, present a vulnerability when a target is under sophisticated physical surveillance. A state-sponsored malicious actor can observe a target's routine and, with high precision, estimate the timestamp of key generation, thereby aiding cryptanalysis. The reliance on predictable or reproducible entropy sources weakens the overall security posture, especially in scenarios where high-value targets are subjected to advanced persistent threats.

Summary of the Invention

The present invention introduces a novel method and system for enhancing cryptographic key generation by leveraging real-time, high-fidelity device orientation and geographical location data as primary entropy sources. This approach transforms inherently volatile physical parameters into robust, unpredictable seeds for cryptographic operations.

The system integrates a GNSS (or GPS) receiver for precise geographical positioning and a digital compass or IMU for accurate device orientation relative to Magnetic North. These sensors provide continuous streams of data that are difficult for an external adversary to measure with sufficient precision to aid cryptanalysis, even with direct physical observation of the target. The acquired location and orientation data are then fed into a cryptographic hashing function, along with other potential entropy sources, to generate a unique encryption seed. The high variability and low latency of these measurements make the resulting seeds exceptionally resistant to prediction and brute-force attacks, offering a significant security advantage over conventional methods.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram of the system for generating encryption seeds using geographical and orientation data.

FIG. 2 is a flowchart illustrating the method for deriving encryption seeds from device location and orientation.

Detailed Description of the Invention

Referring to **FIG. 1**, the system (100) for generating geographical and orientation-based encryption seeds is shown. A mobile device (110) contains a GNSS/GPS Receiver (120) and a Digital Compass/IMU (130). The GNSS/GPS Receiver provides precise Geographical Location Data (121), and the Digital Compass/IMU provides Device Orientation Data (131). These data streams are fed into an Entropy Collection Module (140), which combines them and potentially other entropy sources. The output of the Entropy Collection Module is then passed to a Cryptographic Hashing Function (150), which produces a unique Encryption Seed (160). This seed is then used for cryptographic operations (170).

Referring to **FIG. 2**, the method (200) for deriving encryption seeds is illustrated. The process begins with 'Acquire Geographical Location Data (210)' and 'Acquire Device Orientation Data (220)' simultaneously. These two steps lead to a 'Combine Location and Orientation Data (230)' step. The combined data is then used in a 'Apply Cryptographic Hash Function (240)' step. The output of the hashing function is the 'Generate Encryption Seed (250)', which is then used for 'Cryptographic Operations (260)'. This method ensures that the encryption seed is dynamic and difficult to predict.

Claims

What is claimed is:

1. A method for generating a cryptographic key seed, comprising:

    - acquiring, by a device, real-time geographical location data from a Global Navigation Satellite System (GNSS/GPS) receiver;

- acquiring, by the device, real-time device orientation data from an inertial measurement unit (IMU) or a digital compass;
- combining the acquired geographical location data and device orientation data to form a combined entropy input; and
- applying a cryptographic hash function to the combined entropy input to generate a cryptographic key seed.

2. The method of claim 1, wherein the geographical location data comprises precise latitude and longitude coordinates.

3. The method of claim 1, wherein the device orientation data comprises orientation relative to Magnetic North or a 3D spatial orientation.

4. The method of claim 1, further comprising continuously updating the cryptographic key seed based on changes in the geographical location data or device orientation data.

5. The method of claim 1, wherein the cryptographic key seed is used for encrypting communication between the device and another entity.

6. A system for generating a cryptographic key seed, comprising:
   - a GNSS/GPS receiver configured to acquire real-time geographical location data;
   - an IMU or digital compass configured to acquire real-time device orientation data;
   - an entropy collection module communicatively coupled to the GNSS/GPS receiver and the IMU or digital compass, the entropy collection module configured to combine the geographical location data and device orientation data;
   - a cryptographic hashing unit communicatively coupled to the entropy collection module, the cryptographic hashing unit configured to apply a cryptographic hash function to the combined data to generate a cryptographic key seed.

7. The system of claim 6, further comprising a cryptographic module configured to use the generated cryptographic key seed for encryption operations.

Drawings

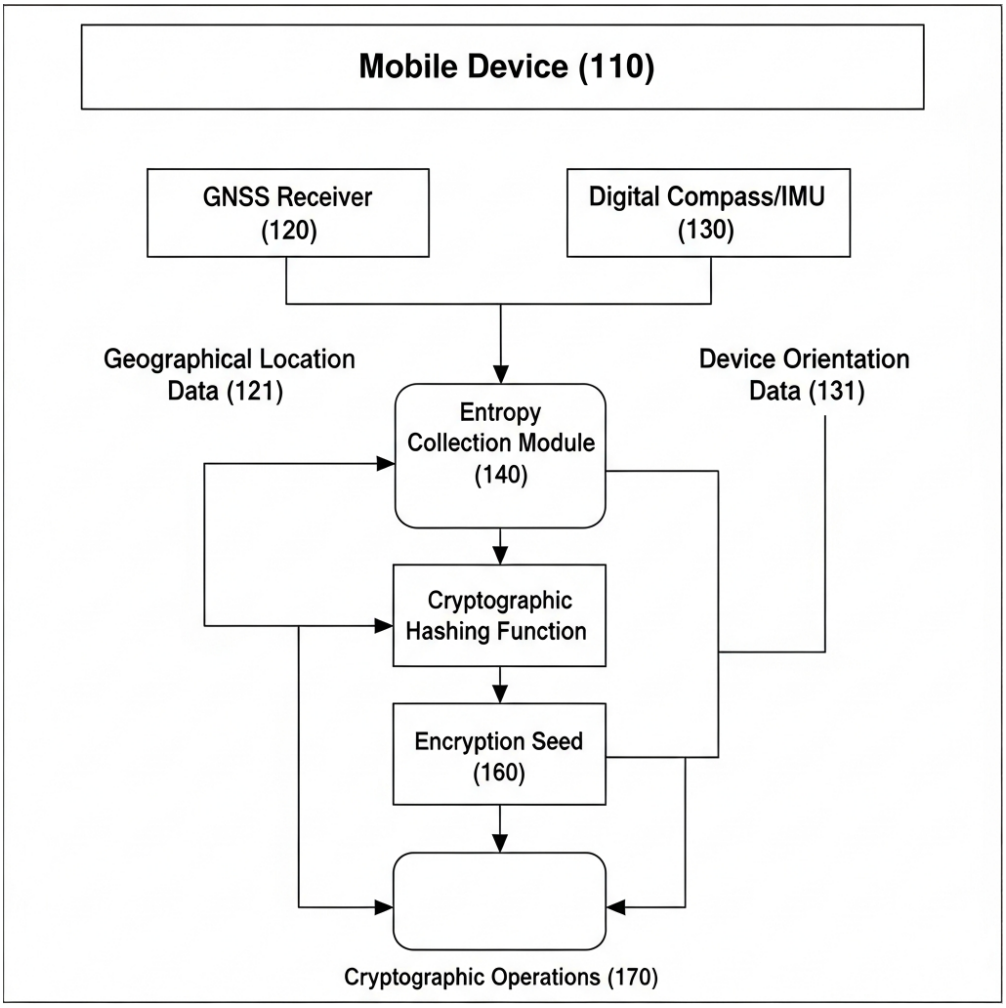


Figure 1: Block diagram of the system for generating encryption seeds using geographical and orientation data.

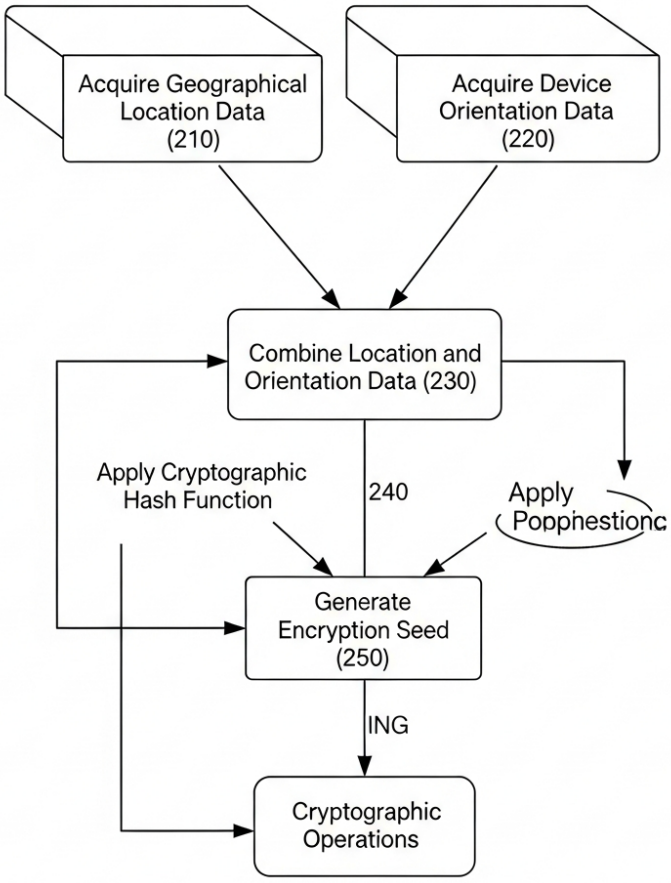


Figure 2: Flowchart illustrating the method for deriving encryption seeds from device location and orientation.